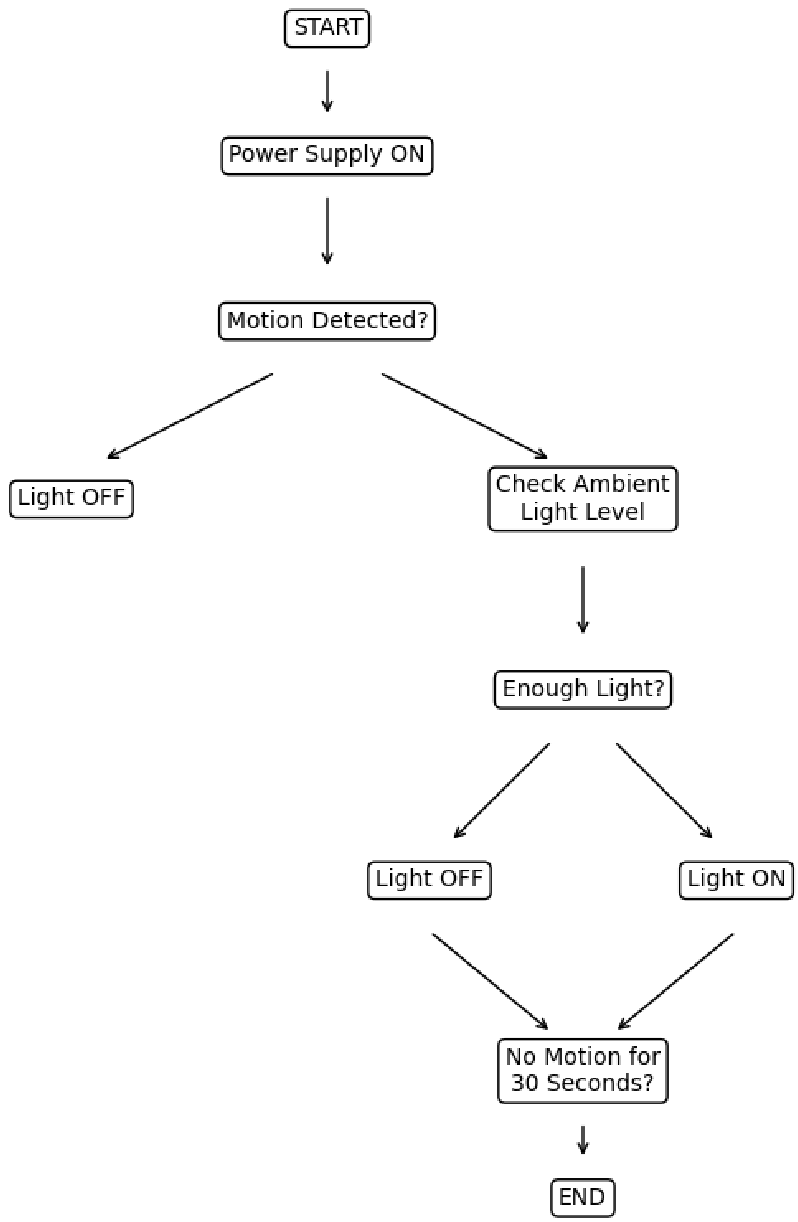
Automation System Design

Smart Home Lighting Automation System

Flowchart



Automation System Design Report

1. Introduction
Automation systems improve convenience, safety, and energy efficiency. A Smart Home Lighting Automation System automatically controls room lighting based on motion detection and ambient light conditions.

2. Objective
To design an automated lighting system that switches lights ON when a person enters a room and switches them OFF when the room is vacant.

3. Components Required
• PIR Motion Sensor
• LDR (Light Dependent Resistor)
• Arduino/ESP32 Microcontroller
• Relay Module
• LED Lamp
• Power Supply
• Connecting Wires

4. Working Principle
The PIR sensor detects motion and sends a signal to the controller. The controller checks ambient light using the LDR. If the room is dark, the relay activates the light. When no movement is detected for 30 seconds, the light automatically turns OFF.

5. Advantages
• Energy Saving
• Automatic Operation
• User Convenience
• Reduced Electricity Consumption
• Reliable and Low Maintenance

6. Applications
• Homes
• Offices
• Hospitals
• Smart Buildings
• Corridors and Parking Areas

7. Conclusion
The Smart Home Lighting Automation System is an efficient and cost-effective solution that minimizes energy wastage while improving comfort and convenience.